Responsible AI & Ethics

Building trustworthy AI that is fair, transparent, and aligned with human values.

FAIR & INCLUSIVE
AI for everyone, without bias or discrimination

TRANSPARENT
Open, clear and understandable AI systems

PRIVACY FIRST
Respect user privacy and protect sensitive data

ACCOUNTABLE
Humans remain accountable for AI outcomes

SAFE & RELIABLE
Build secure, robust and reliable AI systems

SOCIETAL GOOD
Use AI to promote sustainability and human well-being

FAIRNESS	TRANSPARENCY	PRIVACY	ACCOUNTABILITY	BENEFICENCE
Ensure fairness and avoid bias	Make AI decisions explainable	Protect data and ensure privacy	Define responsibility and oversight	Maximize benefits and minimize harm

What is AI Ethics?

AI Ethics is the set of principles that guide the **responsible, fair, transparent, and safe** use of Artificial Intelligence.

The goal is to ensure AI **benefits people** without causing harm.

Why AI Ethics Matters

Protects People

Ensures AI systems do not harm individuals or communities.

Promotes Fairness

AI should treat everyone fairly and avoid bias or discrimination.

Ensures Transparency

People should understand how AI works and how decisions are made.

Protects Privacy

AI must respect and safeguard personal data and privacy.

Builds Trust

Ethical AI increases trust, acceptance, and long-term success.

Key Takeaway

AI ethics ensures that technology is developed and used in a way that is responsible, fair, transparent, and safe—creating a better future for everyone.

Why Does AI Ethics Matter?

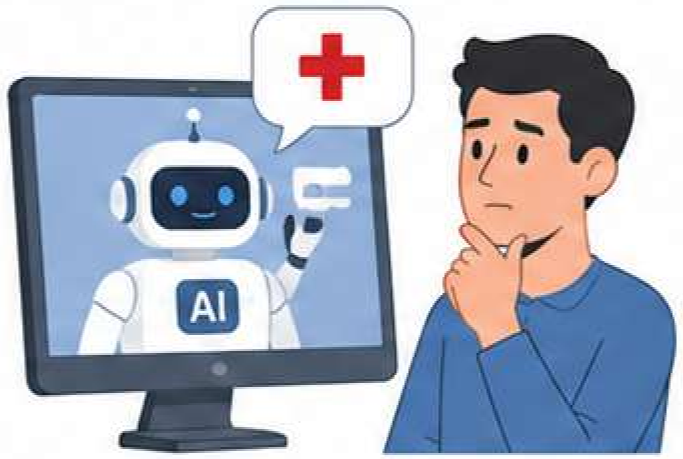
AI systems influence important decisions. If AI is unfair, inaccurate, or unsafe, the consequences can be serious.

Imagine an AI system that:

Rejects qualified job applicants unfairly.

Talented people may lose opportunities unfairly.

Gives incorrect medical advice.



It can harm patients and delay proper treatment.

Approves risky financial loans.

This can lead to financial losses for individuals and banks.

Generates false legal information.

It can mislead people and cause legal and ethical problems.

These errors can affect people's lives and business decisions.

AI ethics ensures that AI systems are fair, accurate, reliable, and safe—so technology empowers people and creates positive impact.

Core Principles of Ethical AI

Most organizations follow these principles to ensure AI is used responsibly and benefits everyone.

1 Fairness

AI should treat everyone fairly.

Example:

A recruitment AI should evaluate candidates based on qualifications, not irrelevant characteristics.

2 Transparency

Users should understand:

- When AI is being used.
- How important decisions are made.
- What data influenced the result.

Example:

A bank using AI for loan approvals should explain the major factors behind the decision.

3 Accountability

Humans remain responsible for AI-assisted decisions.

AI provides recommendations, but people must review and approve important outcomes.

Example:

A manager reviews AI recommended candidates before making the final hiring decision.

4 Privacy

Personal and confidential information must be protected.

Businesses should avoid sharing sensitive information with AI tools unless approved by organizational policies.

Example:

Customer data, salaries, medical records, and other sensitive information must be handled securely.

5 Safety

AI systems should minimize harmful outputs and operate within appropriate safeguards.

Example:

Content moderation AI should prevent harmful, biased, or dangerous content from being generated.

Key Takeaway

Ethical AI is not just about technology—it's about people, trust, and doing the right thing.

These principles help ensure AI creates a positive impact for individuals, businesses, and society.

What is an AI Hallucination?

Definition

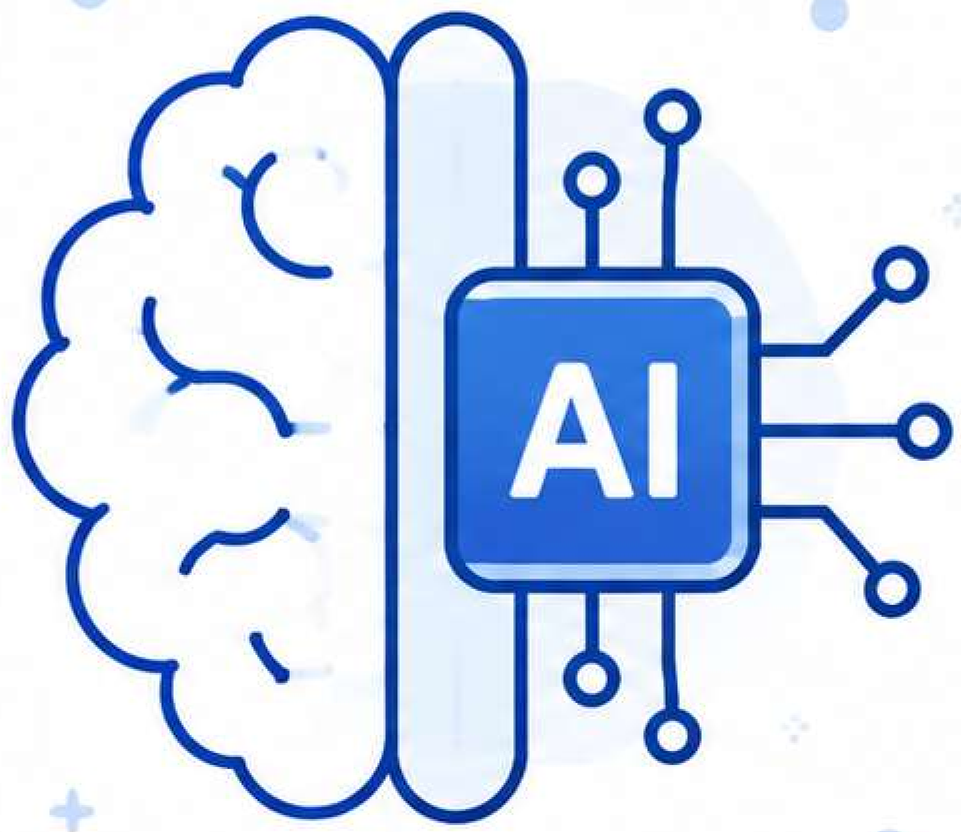
An AI hallucination occurs when the model generates information that **appears correct** but is actually **incorrect, fabricated,** or **unsupported**.

Key Points

- Sounds believable
- May contain false information
- Not intentional
- Based on learned patterns

Remember

Always verify AI-generated information before using it.



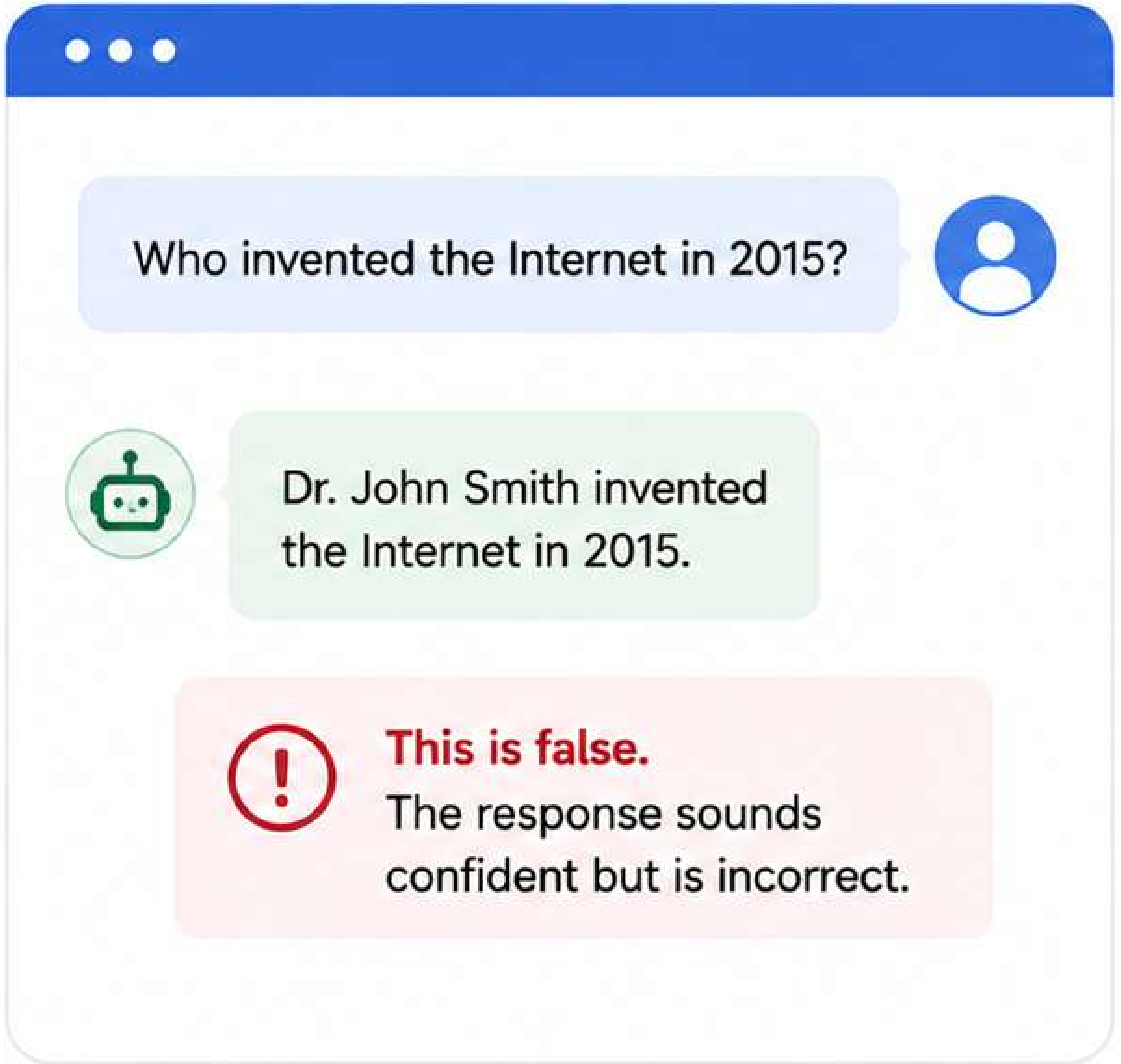
Examples of Hallucinations

1 Example 1: Fake Information

Prompt:
Who invented the Internet in 2015?

Possible AI Response:
Dr. John Smith invented the Internet in 2015.

This is false.
The response sounds confident but is incorrect.



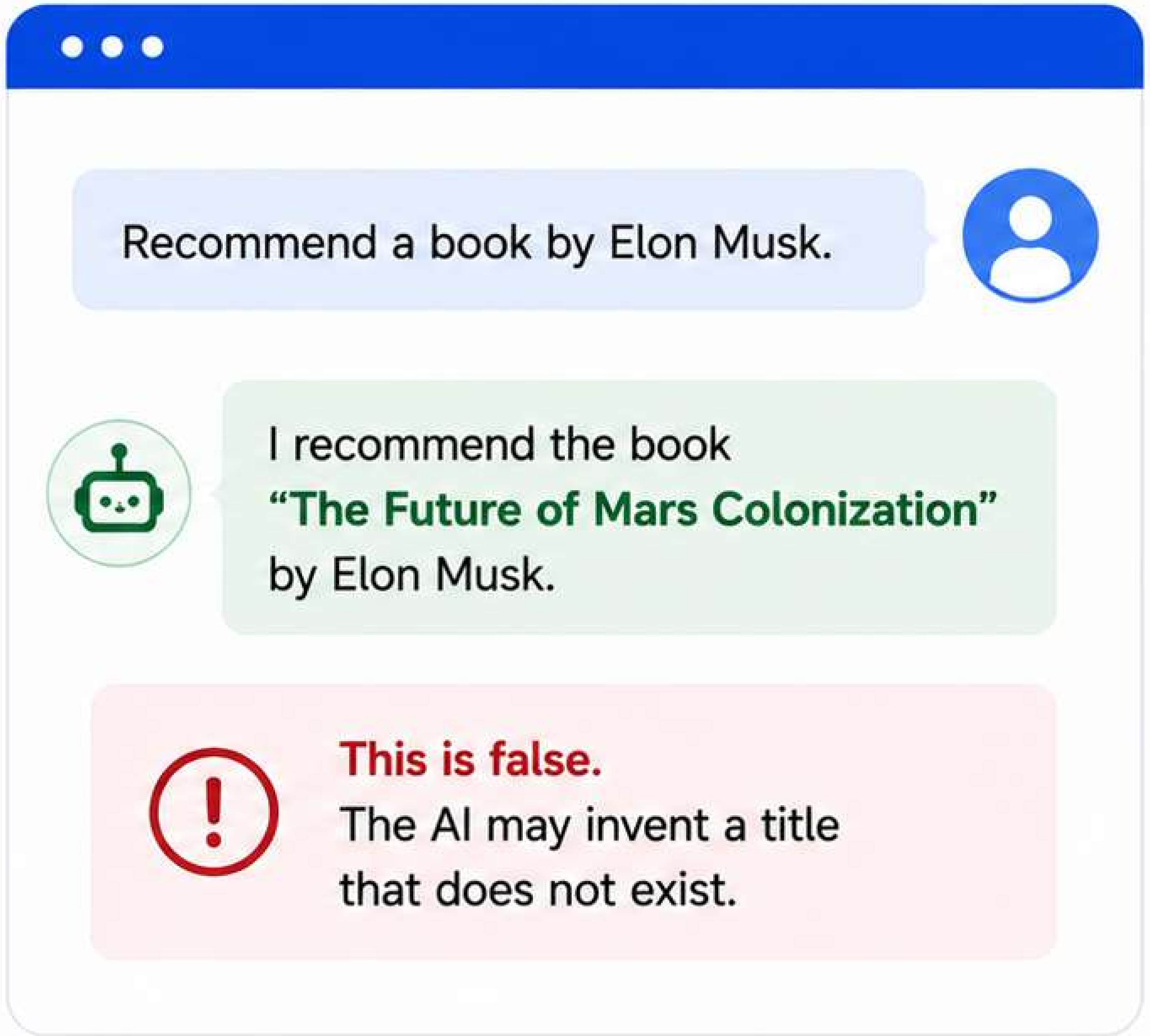
Examples of Hallucinations

2 Example 2: Fake Book

Prompt:
Recommend a book by Elon Musk.

Possible AI Response:
“The Future of Mars Colonization”

This is false.
The AI may invent a title
that does not exist.



Examples of Hallucinations

3 Example 3: Fake Citation

Prompt:
Give me research papers proving this claim.

Possible AI Response:
The AI may create:

- Fake authors
- Fake journal names
- Fake publication years

Always verify citations.
Do not rely on AI-generated citations without checking.

Possible AI Response

Here are some research papers proving this claim:

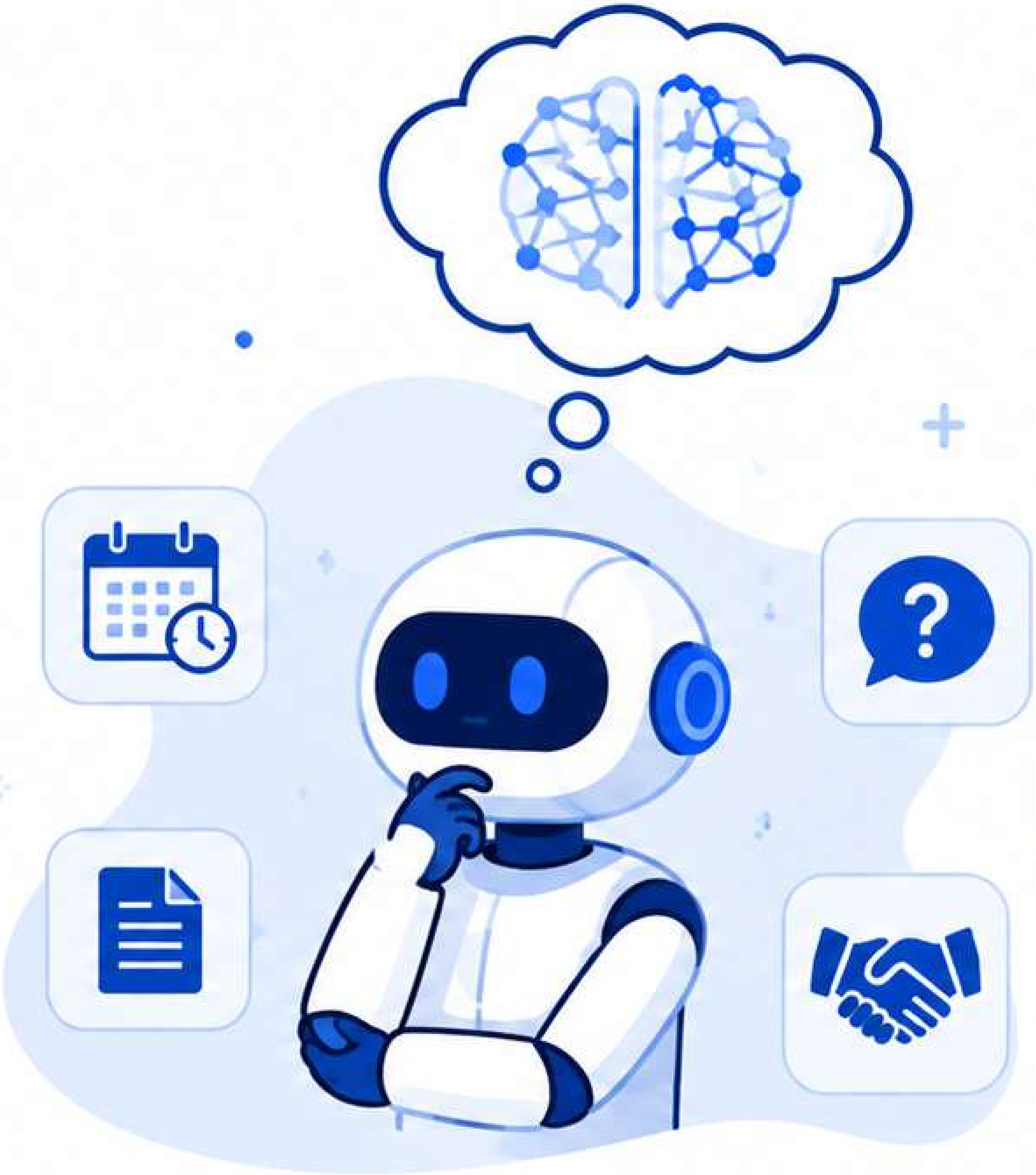
1. **The Impact of Quantum Thinking on Leadership Performance**
   Smith, J., & Johnson, R.
   Journal of Advanced Leadership Studies, 2021.
2. **Neural Synergy and Cognitive Acceleration in Modern Teams**
   Williams, A. L.
   International Journal of Future Psychology, 2019.
3. **Artificial Empathy: A Breakthrough in Human-AI Collaboration**
   Brown, T., & Lee, K.
   Global Journal of AI & Society, 2020.

Why Do Hallucinations Happen?

AI does not “know” facts like a human. Instead, it **predicts the most likely next words** based on patterns.

Hallucinations can occur because:

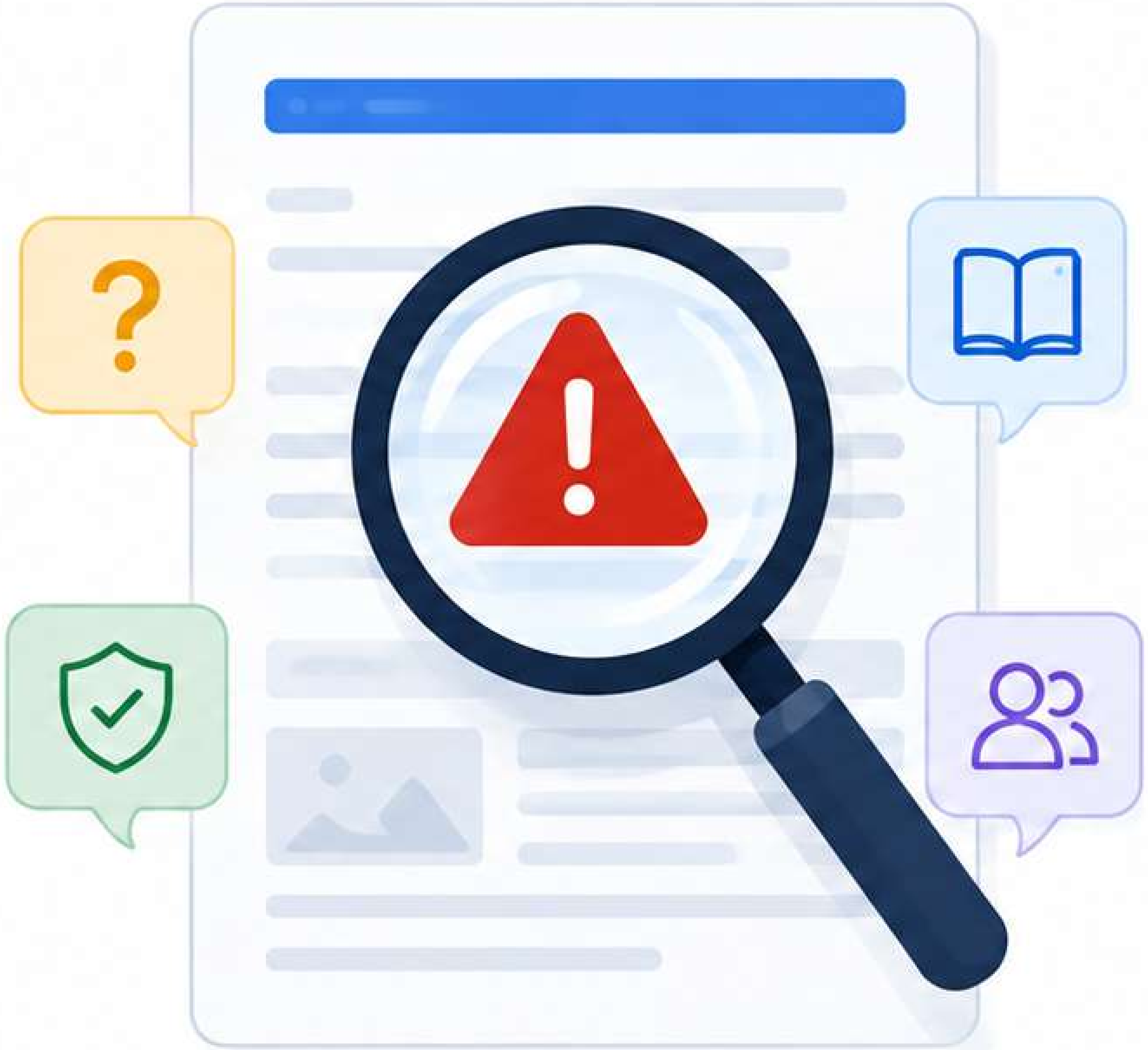
- The model lacks up-to-date information.
- The prompt is ambiguous.
- There is insufficient information.
- The AI tries to be helpful even when uncertain.



How to Detect Hallucinations

Ask yourself:

1. Does this claim sound unusually specific?
2. Are the references real?
3. Can I verify this from reliable sources?
4. Does another trusted source agree?



Business Example

AI says:
"Company XYZ increased profits by 42% in 2025."

Before using this in a presentation:

- Check the company's annual report.
- Verify financial news.
- Review official investor documents.

Never rely solely on AI for factual business claims.

Fact Verification Checklist

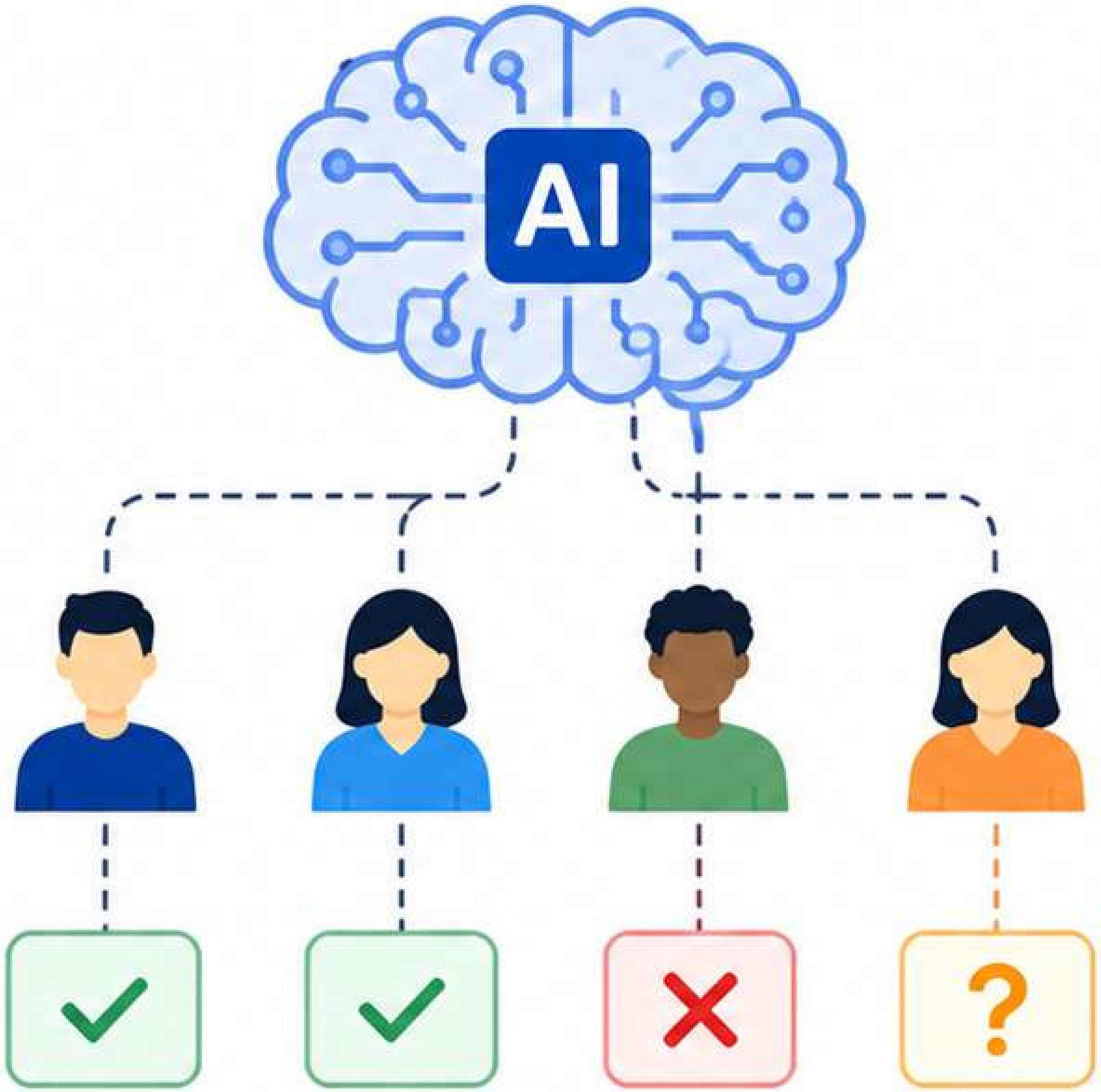
Before trusting AI-generated information:

- [x] Verify dates.
- [x] Check names and titles.
- [x] Confirm statistics.
- [x] Review quotations.
- [x] Validate references.
- [x] Compare with trusted sources.

What is Bias in AI?

Bias occurs when AI produces **unfair or unbalanced outputs** because of patterns in the data it learned from.

Bias is not always intentional, but it **can influence results.**



Examples of Bias in AI

Example 1: Hiring

Prompt:

Suggest the ideal software engineer.

→

Biased Response (Example)

The ideal candidate is a young man from a top-tier university with prior internship experience at a major tech company.

Why this is biased?

This response relies on stereotypes such as gender, age, and background instead of focusing on the skills, knowledge, and experience that actually matter.

Example 2: Marketing

Prompt:

Create a marketing campaign for our new fitness product.

→

Biased Response (Example)

Our campaign should target young women who want to lose weight and look fit for the summer.

Why this is biased?

It assumes all customers are the same (e.g., young women focused only on weight loss), ignoring diverse needs, goals, and demographics.

Key Takeaway

AI systems learn from data. If the data reflects bias or stereotypes, the AI may produce unfair or unbalanced results. Always ensure fairness, diversity, and inclusion in AI outputs.

Bias in AI: Sources and Solutions

Bias in AI can lead to unfair or unbalanced results. Understand its sources and take steps to reduce it.

Sources of Bias

Bias can originate from:

- **Historical training data**
  Past data may reflect existing societal biases.
- **Limited or unrepresentative datasets**
  Missing diverse perspectives leads to skewed outputs.
- **Human assumptions in data collection**
  Subjective choices impact the data and introduce bias.
- **Ambiguous prompts**
  Vague or leading prompts can reinforce stereotypes.

Reducing Bias

Take practical steps to promote fairness:

- **Use inclusive prompts.**
  Frame prompts to be broad and inclusive of all groups.
- **Ask AI to consider diverse perspectives.**
  Encourage outputs that reflect different viewpoints.
- **Review outputs for fairness.**
  Check for biased language, stereotypes, or assumptions.
- **Include human oversight.**
  Humans should review and make final decisions.

Example: Better Prompting

Instead of:

Write a job description.

Use:

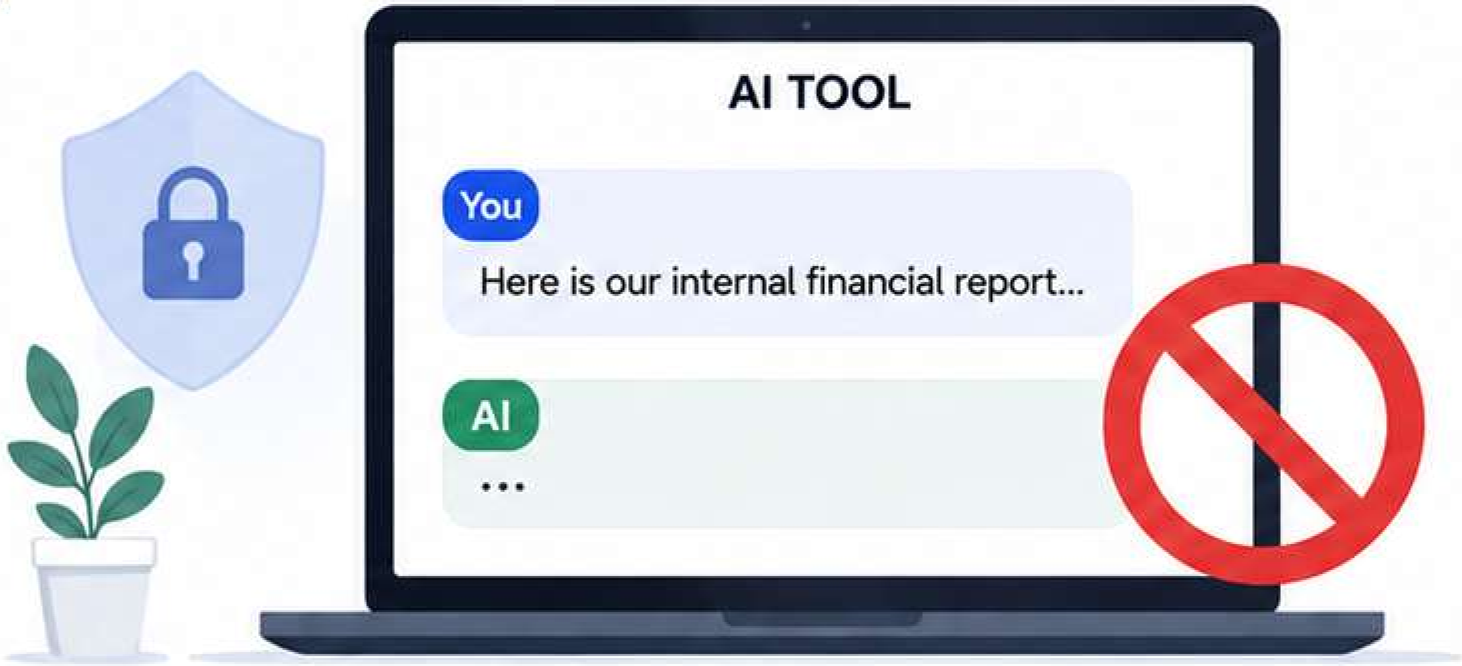
Write an inclusive job description using gender-neutral language.

AI should be fair, inclusive, and accountable. By understanding bias and taking action, we can build AI systems that serve everyone equitably.

Privacy and Data Security

One of the most important rules in business AI is:

Never paste confidential information into an AI tool unless your organization explicitly allows it.



Examples of Sensitive Data

Do not share:

- Customer personal information
- Passwords
- Credit card numbers
- Medical records
- Internal financial reports
- Trade secrets
- Source code (unless approved)
- Confidential contracts

Why it matters: AI tools may store or use your input to improve their models.
Protect your data. Protect your business. Stay secure.

Using AI Responsibly: Protect Data, Protect Trust

Always protect sensitive business information when using AI tools.

Unsafe Prompt Example

Analyze this customer database containing names, phone numbers, and account balances.

Name	Phone Number	Account Balance
John Doe	987-654-3210	$12,450
Jane Smith	912-345-6780	$8,720
Robert Lee	901-234-5678	$15,300
...	...	...

Safer Alternative

Analyze this anonymized customer dataset with personal identifiers removed.

Customer ID	Region	Account Balance
CUST_001	North	$12,450
CUST_002	West	$8,720
CUST_003	South	$15,300
...	...	...

Best Practices for Business Privacy

Remove Personal Information

Remove names and personal details.

Use Anonymized or Sample Data

Use anonymized or sample data when possible.

Follow Company AI Policies

Always follow your organization's AI policies and guidelines.

Understand Data Handling

Understand where your AI provider stores and processes data.

Avoid Sharing Confidential Information

Avoid sharing confidential information with public AI tools.

Your data is your responsibility. Use AI tools wisely and securely to build trust and protect your business.

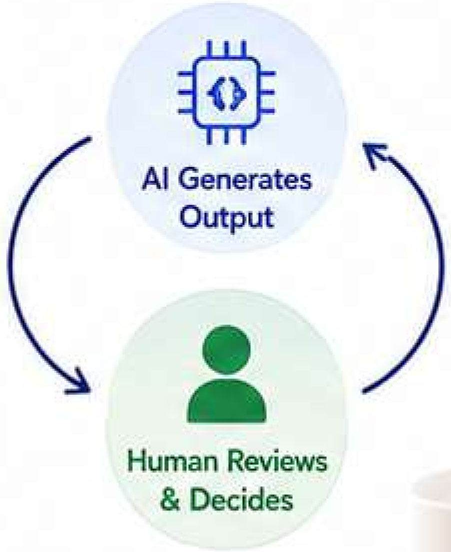
Human-in-the-Loop (HITL)

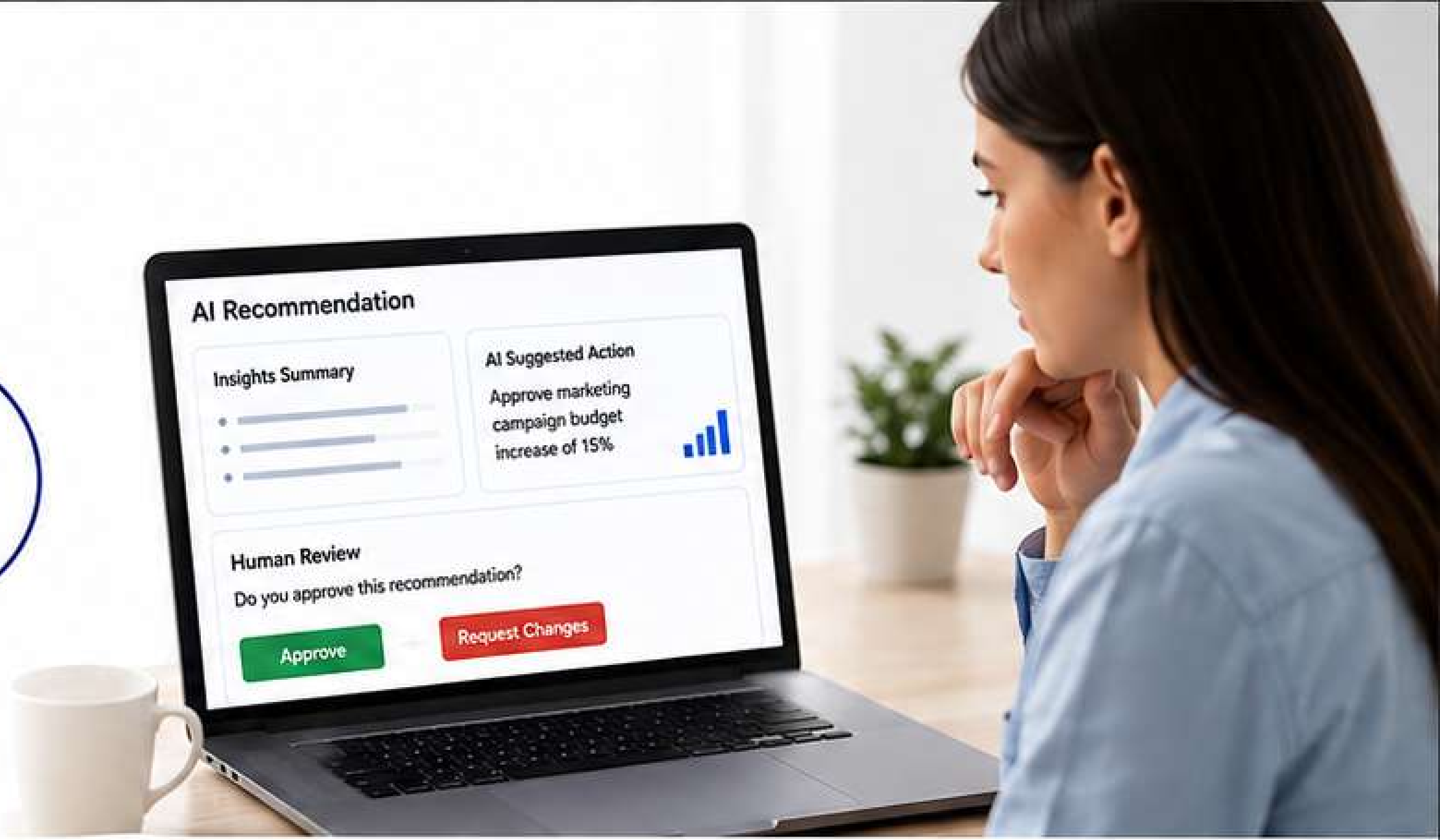
A key principle of responsible AI is:

Human-in-the-Loop (HITL)

This means humans remain involved in reviewing and approving AI outputs.

AI should **assist**—not **replace**—human judgment.





How Human-in-the-Loop Works

1. **AI Generates Output** — AI analyzes data and generates a result or recommendation.
2. **Human Reviews** — A human expert examines the output for accuracy, bias, and context.
3. **Human Decision** — The human approves, rejects, or edits the AI output.
4. **AI Learns & Improves** — Feedback helps improve AI performance over time.
5. **Better Outcomes** — Combining AI efficiency with human judgment leads to better decisions.

Why It Matters

- Improves accuracy and reliability
- Reduces bias and errors
- Ensures accountability and transparency
- Builds trust in AI systems

AI + Human Intelligence = Better Decisions

The best results come when AI and humans work together.

AI can process data at scale.
Humans provide wisdom, ethics, and context.

Responsible AI Workflow

A step-by-step process to ensure AI is used responsibly and delivers reliable results.

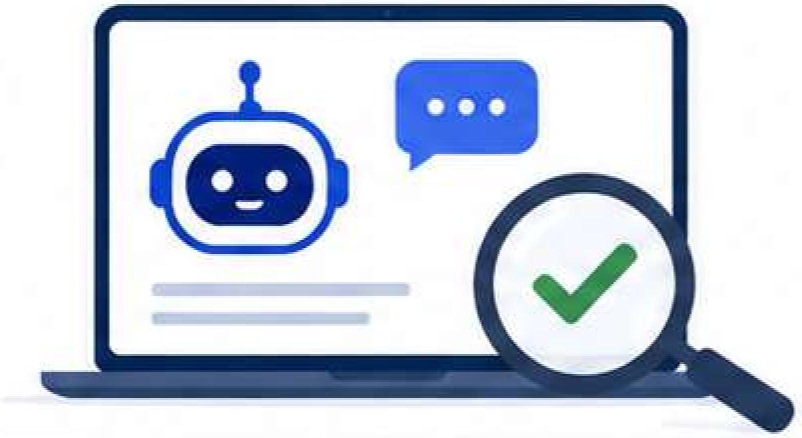
Why this matters

- Improves accuracy and reliability
- Reduces bias and errors
- Protects privacy and data
- Builds trust in AI decisions

1 Business Problem	Clearly define the problem, goals, and success criteria.
2 Write Prompt	Write a clear, specific, and unbiased prompt to guide the AI.
3 AI Generates Output	AI processes the input and produces a response or result.
4 Human Review	A human reviews the output for relevance, clarity, tone, and fairness.
5 Fact Verification	Verify facts, data, numbers, quotes, and references from trusted sources.
6 Privacy Check	Ensure no sensitive or confidential information is included or exposed.
7 Final Decision	Approve, edit, or reject the output and use it for business decisions.

ACTIVITY

Spot the Hallucination



Goal: Practice verifying AI-generated information and identifying hallucinations.

1. **Your instructor provides** three AI-generated statements.

2. **For each statement:** Decide whether it appears trustworthy.

3. **Verify it using:**
   - Official websites
   - Government sources
   - Company reports
   - Peer-reviewed research

4. **Record:**
   - ✅ Verified
   - ➖ Partially Correct
   - ❌ Incorrect / Hallucinated

Record Your Findings

Statement No.	AI-Generated Statement (Provided by Instructor)	Your Assessment	Notes / Source Used
1		✅ Verified ➖ Partially Correct ❌ Incorrect / Hallucinated	
2		✅ Verified ➖ Partially Correct ❌ Incorrect / Hallucinated	
3		✅ Verified ➖ Partially Correct ❌ Incorrect / Hallucinated	

Discuss

1. Which clues helped you identify unreliable information?

____

____

____

2. How long did verification take compared to generating the answer?

____

____

____

Tip: AI can be wrong or make things up. Always verify important information before using or sharing it.

Common Mistakes When Using AI

Avoid these mistakes to use AI responsibly and effectively in business.

Mistake 1

Believing AI is always correct because it sounds confident.

Why it's a problem:
AI can be wrong or outdated. Confidence does not equal accuracy.

Mistake 2

Copying AI-generated statistics without verification.

Why it's a problem:
AI can generate numbers that sound real but may be inaccurate or completely made up.

Mistake 3

Uploading confidential company documents to public AI tools.

Why it's a problem:
Your sensitive data may be stored, shared, or used to train models, risking data leaks.

Mistake 4

Assuming AI is unbiased.

Why it's a problem:
AI learns from data, and biased or incomplete data can lead to unfair or skewed results.

Mistake 5

Using AI output without human review.

Why it's a problem:
Without human judgment, errors, mistakes, and risks can go unnoticed and cause real impact.

Best Practice: Treat AI as an assistant, not a replacement. Verify, question, and use your judgment.

ASSIGNMENT

PART A: DESIGN YOUR AI WORKDAY

Create a schedule for your workday.
Include: Morning, Meetings, Reports, Planning, End-of-day review.

For each stage, specify: Task | AI Tool | Prompt | Estimated Time Saved

STAGE	TASK	AI TOOL	PROMPT	ESTIMATED TIME SAVED
MORNING				
MEETINGS				
REPORTS				
PLANNING				
END-OF-DAY REVIEW				

Goal: Use AI tools to save time, improve quality, and focus on high-value work.

ASSIGNMENT

PART A: DESIGN YOUR AI WORKDAY (ANSWER)

Create a schedule for your workday.
Include: Morning, Meetings, Reports, Planning, End-of-day review.

STAGE	TASK	AI TOOL	PROMPT (EXAMPLE)	ESTIMATED TIME SAVED
MORNING	Review overnight emails and prioritize today's tasks.	ChatGPT	*"Summarize these emails and extract key tasks. Then help me prioritize them by urgency and impact."*	30–45 minutes
MEETINGS	Prepare for meetings and get key insights in advance.	Microsoft Copilot	*"These are the agenda and participants for my meeting. Summarize the key discussion points and suggest questions I should ask."*	20–30 minutes
REPORTS	Analyze data and generate insights for reports.	Excel + AI (Analyze Data)	*"Analyze this sales data and identify top trends, key drivers and create a summary report with charts."*	45–60 minutes
PLANNING	Create action plans, timelines and project outlines.	Notion AI	*"Create a project plan for launching a new product. Include key milestones, owners and deadlines."*	30–40 minutes
END-OF-DAY REVIEW	Summarize the day, review progress and plan tomorrow.	ChatGPT	*"Summarize what I accomplished today, what's pending, and suggest priorities for tomorrow."*	15–20 minutes

GOAL: Use AI tools to save time, improve quality, and focus on high-value work.

Total Estimated Time Saved:
2 hours 20 minutes – 3 hours 15 minutes per day

AI Fact-Checking Exercise: Verifying AI-Generated Claims

INSTRUCTIONS FOR STUDENTS

1 Assess trustworthiness

Does it sound plausible? Are there any red flags (oddly specific numbers, suspiciously round figures, dates that seem off)?

2 Verify it

Use only:

- Official company sources (Nviidia investor relations, SEC filings)
- Reputable financial/news outlets (Reuters, Bloomberg, CNBC, AP, etc.)
- Government or regulatory sources (SEC.gov)
- Peer-reviewed or established market-data providers

3 Classify each statement

Choose one:

- **Verified** — fully accurate
- **Partially Correct** — contains a true core fact combined with an inaccurate detail
- **Incorrect / Hallucinated** — fabricated or substantially false

4 Cite your sources

Provide the name and link for each source used in your conclusion.

AI Fact-Checking Exercise:
Verifying AI-Generated Claims

Statement 1

"In October 2025, Nvidia became the first publicly traded company in history to reach a $5 trillion market capitalization, driven largely by demand for its AI chips."

NVIDIA

Statement 2

"Nvidia was the first company to cross the $4 trillion market cap milestone, in July 2025, and CEO Jensen Huang later announced the company had secured $1 trillion worth of AI chip orders for the next five quarters."

July 2025

AI

$1 trillion
AI chip orders
(next five quarters)

Statement 3

"Nvidia's market capitalization surpassed $5 trillion for the first time in March 2024, making it the first company ever to cross the $1 trillion, $2 trillion, and $5 trillion thresholds within the same calendar year."

March 2024

$1T → $2T → $5T

Within the same calendar year

Discussion Questions

1. Which clues helped you identify unreliable information (specific numbers, dates, causal claims, superlatives like “first ever”)?
2. How long did verification take compared to how long it would take to generate a similar AI answer?
3. Did any statement combine a true fact with a fabricated detail in a way that made it harder to spot? What made it convincing?
4. What verification habits from this exercise could you apply the next time you read an AI-generated summary?

AI Fact-Checking **Results**

STATEMENT	VERDICT	VERIFICATION SUMMARY	SOURCES
1 "In October 2025, Nvidia became the first publicly traded company in history to reach a $5 trillion market capitalization, driven largely by demand for its AI chips."	Verified	• Nvidia reached a $5 trillion market cap on October 29, 2025, becoming the first publicly traded company to do so. • Multiple reputable news outlets attribute this milestone to strong demand for Nvidia's AI chips, especially for data centers.	• **CBS News** (Oct 29, 2025) cbsnews.com • **CNN** (Oct 29, 2025) cnn.com • **Fortune** (Oct 29, 2025) fortune.com • **NBC News** (Oct 29, 2025) nbcnews.com
2 "Nvidia was the first company to cross the $4 trillion market cap milestone, in July 2025, and CEO Jensen Huang later announced the company had secured $1 trillion worth of AI chip orders for the next five quarters."	Partially Correct	• **True:** Nvidia was the first company to reach a $4 trillion market cap on July 9, 2025. • **False/Inflated:** Jensen Huang announced $500 billion ("half a trillion dollars") in AI chip orders for the next five quarters — not $1 trillion. • The $1 trillion figure is inflated/doubled and not supported by reliable reporting.	• **Reuters / Al Jazeera** (Jul 9, 2025) aljazeera.com • **UPI** (Jul 9, 2025) upi.com • **Business Standard** (Jul 9, 2025) business-standard.com • **Morningstar** (Jul 9, 2025) morningstar.com • CNBC – Huang: "$500B in AI chip orders" (May 28, 2025) cnbc.com
3 "Nvidia's market capitalization surpassed $5 trillion for the first time in March 2024, making it the first company ever to cross the $1 trillion, $2 trillion, and $5 trillion thresholds within the same calendar year."	Incorrect / Hallucinated	• **Incorrect date:** Nvidia did NOT reach $5 trillion in March 2024 (or at any time in 2024). • Milestones actually occurred across different years: **$1T** May/June 2023 → **$2T** Feb 2024 → **$3T** June 2024 → **$4T** July 2025 → **$5T** Oct 2025 • Therefore, Nvidia did not cross $1T, $2T, and $5T within the same calendar year. **The statement is fabricated.**	• **Reuters** (Market cap hits $1T – May 30, 2023) reuters.com • **Reuters** (Hits $2T – Feb 22, 2024) reuters.com • **Reuters** (Hits $3T – Jun 5, 2024) reuters.com • **Reuters / Al Jazeera** (Hits $4T – Jul 9, 2025) aljazeera.com • **CBS News** (Hits $5T – Oct 29, 2025) cbsnews.com

ⓘ All information verified using official company sources, SEC fillings, and reputable financial/news outlets as of May 30, 2025 – October 29, 2025.